LAPORAN PERANCANGAN BASIS DATA

Sintaks SQL DDL (Create Table) & DML (Insert 5 Data)

PROYEK TUBES: STREAMLY (TIKTOK LIVE AI CHAT ANALYZER)

Implementasi Skema Relasional Persisten 5 Tabel Sesuai Rancangan Asli:
tubes/Streamly/skets/streamly_db_design/schema.sql
Mencakup Sesi Streaming, Riwayat Komentar, Analisis AI, dan Keterlacakan Data (Provenance).

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PROGRAM STUDI S1 INFORMATIKA
FAKULTAS INFORMATIKA — UNIVERSITAS TELKOM

Mata Kuliah: Implementasi Perangkat Lunak (IMPAL)
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1. Deskripsi Skema Basis Data 5 Tabel Streamly

Rancangan basis data relasional ini mengacu secara langsung pada dokumen sketsa asli di tubes/Streamly/skets/streamly_db_design/schema.sql dan ERD.md. Di dalam sistem Streamly, Redis bertugas mengelola antrean pesan waktu nyata (*real-time queue*) dan cache insight cepat, sedangkan MySQL bertindak sebagai penyimpan riwayat permanen (*durable history*).

Skema terdiri dari **5 tabel utama** yang saling terhubung melalui relasi kunci asing (Foreign Key):

No	Nama Tabel	Peran & Kardinalitas	Deskripsi Fungsi
1	**live_sessions**	1 baris per siaran live	Mencatat satu siklus monitoring scraper terhadap live streaming seorang kreator TikTok.
2	**messages**	N baris per session (1:N)	Menyimpan rekaman obrolan mentah penonton TikTok yang ditangkap scraper.
3	**insights**	N baris per session (1:N)	Menyimpan hasil analisis AI Gemini dalam 3 jendela waktu (5m, 20m, 45m).
4	**insight_messages**	Many-to-Many (M:N)	Tabel relasi pelacakan (provenance) komentar mentah mana saja yang dianalisis pada laporan 5m.
5	**insight_dependencies**	Many-to-Many (M:N)	Tabel relasi turunan hierarkis (laporan 5m menjadi input laporan 20m, dan laporan 20m menjadi input 45m).

2. Sintaks SQL DDL (Create 5 Tabel)

Berikut adalah sintaks DDL (Data Definition Language) lengkap untuk membuat basis data streamly_db dan kelima tabel relasional persisten menggunakan MySQL 8.0+ sesuai berkas rancangan:

```
-- ============================================================================
-- DDL PEMBUATAN DATABASE DAN 5 TABEL UTAMA STREAMLY
-- Rujukan: tubes/Streamly/skets/streamly_db_design/schema.sql
-- ============================================================================
CREATE DATABASE IF NOT EXISTS streamly_db
  CHARACTER SET utf8mb4 COLLATE utf8mb4_0900_ai_ci;
USE streamly_db;

-- 1. TABEL SESI SIARAN LANGSUNG (live_sessions)
CREATE TABLE IF NOT EXISTS live_sessions (
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
  username VARCHAR(100) NOT NULL COMMENT 'TikTok creator username being monitored',
  room_id VARCHAR(100) NULL COMMENT 'TikTok room ID when provided by connector',
  status ENUM('starting','active','ended','failed') NOT NULL DEFAULT 'starting',
  started_at DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  ended_at DATETIME(3) NULL,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  KEY idx_sessions_username_started (username, started_at),
  KEY idx_sessions_status (status),
  CONSTRAINT chk_session_times CHECK (ended_at IS NULL OR ended_at >= started_at)
) ENGINE=InnoDB;

-- 2. TABEL KOMENTAR CHAT AUDIENS (messages)
CREATE TABLE IF NOT EXISTS messages (
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
  session_id BIGINT UNSIGNED NOT NULL,
  message_time DATETIME(3) NOT NULL,
  tiktok_user_id VARCHAR(100) NULL COMMENT 'TikTok commenter unique ID if available',
  display_name VARCHAR(100) NULL,
  message TEXT NOT NULL,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  KEY idx_messages_session_time (session_id, message_time),
  CONSTRAINT fk_messages_session FOREIGN KEY (session_id)
    REFERENCES live_sessions(id) ON DELETE CASCADE
) ENGINE=InnoDB;

-- 3. TABEL LAPORAN ANALISIS AI (insights)
CREATE TABLE IF NOT EXISTS insights (
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
  session_id BIGINT UNSIGNED NOT NULL,
  batch_type ENUM('5m','20m','45m') NOT NULL,
  sentiment_score DECIMAL(5,4) NULL COMMENT 'Expected range -1.0000 to 1.0000',
  sentiment_trend ENUM('rising','falling','stable') NULL,
  emerging_topics JSON NULL,
  summary TEXT NOT NULL,
  actionable_insight TEXT NOT NULL,
  message_count INT UNSIGNED NOT NULL DEFAULT 0,
  model_name VARCHAR(100) NULL,
  generated_at DATETIME(3) NOT NULL,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  KEY idx_insights_session_type_time (session_id, batch_type, generated_at),
  CONSTRAINT fk_insights_session FOREIGN KEY (session_id)
    REFERENCES live_sessions(id) ON DELETE CASCADE,
  CONSTRAINT chk_sentiment_score CHECK (sentiment_score IS NULL OR (sentiment_score >= -1 AND sentiment_score <= 1))
) ENGINE=InnoDB;

-- 4. TABEL PROVENANCE PESAN KE INSIGHT (insight_messages)
CREATE TABLE IF NOT EXISTS insight_messages (
  insight_id BIGINT UNSIGNED NOT NULL,
  message_id BIGINT UNSIGNED NOT NULL,
  PRIMARY KEY (insight_id, message_id),
  KEY idx_insight_messages_message (message_id),
  CONSTRAINT fk_im_insight FOREIGN KEY (insight_id)
    REFERENCES insights(id) ON DELETE CASCADE,
  CONSTRAINT fk_im_message FOREIGN KEY (message_id)
    REFERENCES messages(id) ON DELETE CASCADE
) ENGINE=InnoDB;
```

```
-- 5. TABEL RELASI DEPENDENSI INSIGHT HIERARKIS (insight_dependencies)
CREATE TABLE IF NOT EXISTS insight_dependencies (
  insight_id BIGINT UNSIGNED NOT NULL,
  source_insight_id BIGINT UNSIGNED NOT NULL,
  PRIMARY KEY (insight_id, source_insight_id),
  KEY idx_insight_dependencies_source (source_insight_id),
  CONSTRAINT fk_id_insight FOREIGN KEY (insight_id)
    REFERENCES insights(id) ON DELETE CASCADE,
  CONSTRAINT fk_id_source_insight FOREIGN KEY (source_insight_id)
    REFERENCES insights(id) ON DELETE CASCADE,
  CONSTRAINT chk_insight_not_self CHECK (insight_id <> source_insight_id)
) ENGINE=InnoDB;
```

3. Sintaks SQL DML (Insert 5 Data Tiap Tabel)

Berikut adalah sintaks DML (Data Manipulation Language) yang memasukkan tepat **5 baris data contoh** ke dalam masing-masing dari 5 tabel (total 25 baris data). Data disusun realistis sesuai skenario operasional siaran langsung TikTok di Indonesia dan menjaga integritas referensial kunci asing.

```
-- ============================================================================
-- DML INSERT 5 DATA CONTOH UNTUK MASING-MASING DARI 5 TABEL
-- ============================================================================

-- 1. INSERT 5 DATA live_sessions
INSERT INTO live_sessions (id, username, room_id, status, started_at, ended_at) VALUES
(1, 'glowup_official', 'ROOM_7382910481920', 'ended', '2026-10-19 19:00:00.000', '2026-10-19 21:00:00.000'),
(2, 'urbanstreet.id', 'ROOM_7382910481921', 'active', '2026-10-19 20:30:00.000', NULL),
(3, 'dapurbunda_live', 'ROOM_7382910481922', 'ended', '2026-10-18 14:00:00.000', '2026-10-18 15:45:00.000'),
(4, 'gadgetkita_shop', 'ROOM_7382910481923', 'ended', '2026-10-18 19:30:00.000', '2026-10-18 20:30:00.000'),
(5, 'batikheritagemodern', 'ROOM_7382910481924', 'active', '2026-10-19 21:00:00.000', NULL);

-- 2. INSERT 5 DATA messages
INSERT INTO messages (id, session_id, message_time, tiktok_user_id, display_name, message) VALUES
(1, 1, '2026-10-19 19:07:12.150', 'tt_user_8819', 'Rina Pratiwi', 'Kak, serum etalase 1 aman buat kulit sensitif bruntusan gak?'
(2, 1, '2026-10-19 19:08:45.300', 'tt_user_2234', 'Budi Santoso', 'Spill diskonnya dong kak, kalau beli 2 dapet gratis ongkir ka
(3, 1, '2026-10-19 19:09:10.820', 'tt_user_9941', 'Dewi Lestari', 'Udah checkout etalase 1 kak! Tolong kirim malam ini juga yaa'
(4, 1, '2026-10-19 19:10:02.410', 'tt_user_5512', 'Agus Setiawan', 'Kak suaranya kok agak kresek-kresek ya di live barusan?'),
(5, 1, '2026-10-19 19:11:15.600', 'tt_user_7733', 'Maya Sari', 'Serumnya ada sertifikat BPOM dan Halal gak kak?');

-- 3. INSERT 5 DATA insights
INSERT INTO insights (id, session_id, batch_type, sentiment_score, sentiment_trend, emerging_topics, summary, actionable_insight
(1, 1, '5m', 0.8200, 'stable', '["kulit sensitif", "diskon beli 2", "bpom"]', 'Penonton antusias menanyakan kecocokan serum dan
(2, 1, '5m', 0.7500, 'falling', '["audio kresek", "packing aman", "spill tekstur"]', 'Terdapat keluhan minor mengenai kejelasan
(3, 1, '5m', 0.8900, 'rising', '["promo bundle", "free gift", "checkout sukses"]', 'Lonjakan sentimen positif terjadi berkat bon
(4, 1, '20m', 0.8200, 'rising', '["rekomendasi skincare", "flash sale gajian", "kecepatan kirim"]', 'Evolusi obrolan menunjukkan
(5, 1, '45m', 0.8450, 'stable', '["brand trust", "retensi tinggi", "sold out etalase 1"]', 'Secara umum sesi live berjalan efekt

-- 4. INSERT 5 DATA insight_messages (PROVENANCE KOMENTAR KE INSIGHT 5m)
INSERT INTO insight_messages (insight_id, message_id) VALUES
(1, 1),
(1, 2),
(1, 3),
(1, 4),
(1, 5);

-- 5. INSERT 5 DATA insight_dependencies (SINTESIS HIERARKIS ANTAR-INSIGHT)
INSERT INTO insight_dependencies (insight_id, source_insight_id) VALUES
(4, 1),
(4, 2),
(4, 3),
(5, 4),
(5, 1);
```

4. Pengujian Integritas Relasi Data

Kueri berikut digunakan untuk menguji keterhubungan data antar tabel dalam lingkungan sistem nyata:

A. Kueri Rekap Sesi Live Streaming dan Total Insight yang Dihasilkan:

```
SELECT
    ls.id AS session_id,
    ls.username,
    ls.room_id,
    ls.status,
    COUNT(DISTINCT m.id) AS total_messages,
    COUNT(DISTINCT i.id) AS total_insights
FROM live_sessions ls
LEFT JOIN messages m ON ls.id = m.session_id
LEFT JOIN insights i ON ls.id = i.session_id
WHERE ls.id = 1
GROUP BY ls.id, ls.username, ls.room_id, ls.status;
```

B. Kueri Pelacakan Komentar Mentah yang Mendasari Saran AI (Provenance):

```
SELECT
    i.batch_type,
    i.sentiment_score,
    i.actionable_insight,
    msg.display_name,
    msg.message
FROM insights i
JOIN insight_messages im ON i.id = im.insight_id
JOIN messages msg ON im.message_id = msg.id
WHERE i.id = 1;
```

C. Kueri Keterkaitan Sintesis Hierarkis (Insight 20m & 45m terhadap Insight Sumber):

```
SELECT
    parent.id AS parent_insight_id,
    parent.batch_type AS parent_type,
    parent.summary AS parent_summary,
    source.id AS source_insight_id,
    source.batch_type AS source_type,
    source.summary AS source_summary
FROM insight_dependencies id_dep
JOIN insights parent ON id_dep.insight_id = parent.id
JOIN insights source ON id_dep.source_insight_id = source.id
ORDER BY parent.id, source.id;
```

Kesimpulan Pengujian:
Semua relasi kunci asing (Foreign Key) terhubung secara utuh tanpa anomali data. Skema 5 tabel ini memenuhi kebutuhan fungsional arsitektur Streamly secara presisi, memungkinkan pelacakan penuh dari sesi live, riwayat obrolan penonton, hingga hasil sintesis bertingkat AI.